

CS & IT ENGINEERING



'C' Programming

C - Tokens

Lecture No.- 01



By- Satya sir

Topics to be Covered



'C' Tokens



- Evolution of Languages
- Types of Languages
- Translators
- 'C' Language - features

Let us 'C'
Yeshwant Kanetkar

SYLLABUS



- 'C' Tokens
- Data Types
- I/O Functions
- Control Statements
- Pointers
- Arrays, Strings
- Functions - Storage classes, Recursion, Scope
- Structures, Unions

Excluded

Preprocessor directives
File handling
Enum, bit fields

PDS WEIGHTAGE



YEAR	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015
MARKS	15	11	11	8	12	12	10	14	12	18

Handwritten yellow checkmarks are placed above the years 2024, 2023, 2022, and 2021.

Handwritten yellow notes are placed below the marks for 2024 and 2023:

- 7+8 (under 15)
- 5+6 (under 11)



Computer : An Electronic device, that accepts input, Process input and Produces output, also store results.

(OR)

Combination of Hardware and Software

Functional units :

1) I/O devices (or) I/O unit

2) Processor (CPU)

3) Memory

} H/W

+

S/W

- Operating System
- Translators
- Debugger
- Loader
- Application s/w



Topic : C – Tokens - 1



Software : Set (or) Collection of Programs

Program : Collection of Instructions / Statements Written Using a Programming Language.

Any executable
command

Computer Languages



Low-Level

High-level

Language Directly (or) Easily understandable
by Machine

Also called as
Binary Language
1/0

Machine
Language

Also called as -

Assembly
Language

Symbolic Language
Ex: ADD C, B, A

Easily understandable by User.

C, C++, Java, Python, VB, R,
Pascal, COBOL, Basic - - - - -



TRANSLATORS

– A Program that translates (converts) one form of Language to another form.

– 3 Types of Translators :

a) **Assembler** : Assembly Language to Machine Language : Fastest

Less Effective { b) **Compiler** (C, C++, Java) : High level Language to Machine Language $\Rightarrow$ All code at a time : Faster

More Effective { c) **Interpreter** : High level Language to Machine Language $\Rightarrow$ Single line at a time : slow
(Python)



Topic : C – Tokens - 1



C

- Written by "Dennis Ritchie"

- In year 1972

- at AT&T Bell Labs, U.S.A

- ALGOL → BCPL → B → C



Topic : C – Tokens - 1



Year	Languages / Machine	Invented by
<u>1843</u>	First Machine Algorithm	<u>Ada Lovelace</u>
<u>1940's</u>	<u>Plankalkul</u>	<u>Konrad Zuse</u>
<u>1943</u>	<u>Turing Machine</u>	Alan Turing
<u>1949</u>	<u>Assembly Language</u>	<u>William Schmitt</u>
1952	Autocode	University Of Manchester
1957	Fortran	John Backus
1958	ALGOL, LISP	American and European scientists, John McCarthy
1959	COBOL	Dr. Grace Murray Hopper
1964	BASIC	Bill Gates
1970	PASCAL	Niklaus Wirth
<u>1972</u>	SmallTalk, <u>C</u> , SQL	(Alan Kay, Adele Goldberg, and Dan Ingalls,) Dennis Ritchie, (Raymond Boyce and Donald Chamberlain)



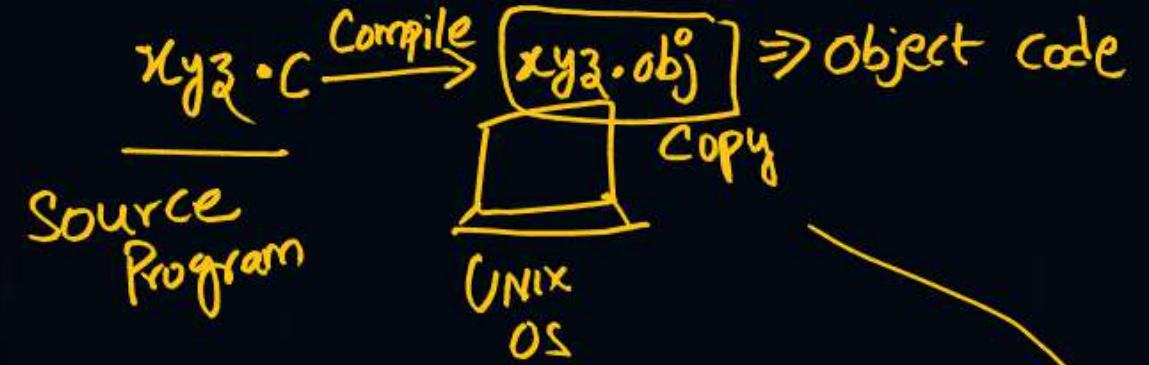
Topic : C – Tokens - 1



Year	Languages / Machine	Invented by
1980s	Ada	Jean Ichbiah's team
1983	C++, Objective C	Bjarne Stroustrup, (Brad Cox and Tom Love)
1987	PERL	Larry Wall
1990	Haskell	Haskell Brooks Curry
1991	Python, VB	Guido Van Rossum, Microsoft
1993	Ruby	Yukihiro Matsumoto
1995	Java, Javascript, PHP	James Gosling, Brendan Eich, Rasmus Lerdorf
2000	C#	Microsoft
2003	Scala, Groovy	Martin Odersky, (James Strachan and Bob McWhirter)
2009	GO (Google)	Google
2014	Swift	Apple



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'C' Features / Characteristics

→ 'C' is General Purpose Prg Language

→ 'C' is Simple, Easy to learn, understand, Use

→ 'C' is Robust

→ 'C' is Procedure-oriented

→ 'C' is Compiled Prg language

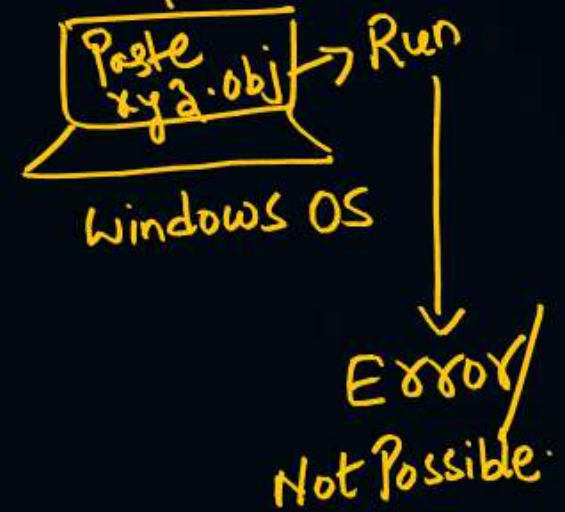
→ 'C' is Portable

→ 'C' is Platform Dependant

→ 'C' is Structured Prg Lang

→ 'C' is middle-level language

[It Suits for both System software
and application software]





2 mins Summary



- Languages Evolution
- Types of languages
- Translators
- 'C' history
- 'C' features

To be cont'd ... 



THANK - YOU